

SANDHYA BARAL

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[LinkedIn](#) | [GitHub](#) | [Personal Website](#)

RESEARCH INTERESTS

Computer Vision · Natural Language Processing · Generative AI · AI for Social Impact
AI for Security · Privacy-Preserving Machine Learning · Interpretable & Transparent AI

EDUCATION

Bachelor in Computer Engineering

Pulchowk Campus, Institute of Engineering, Tribhuvan University

Undergraduate School

October 2021 – June 2025

Academic Standing: First Division — **79.22%**

Enrollment Rank: 286 / 10,000 (**Top 3%**)

- **Relevant Coursework:** Linear Algebra, Calculus, Probability & Statistics, Data Structures & Algorithms, Database Management System, Operating Systems, Artificial Intelligence, Computer Networks, Applied Data Science, Embedded Systems, Digital Signal Analysis and Processing.
- **Thesis:** Handwritten Devanagari **Character Recognition:** Dataset Creation and Benchmarking with Transfer Learning
Thesis Supervisor: [Prof. Sanjeed Prasad Pandey](#)

PUBLICATIONS

- Handwritten Devanagari Character Recognition: Conjoint Consonants Dataset Creation and Benchmarking with Transfer Learning. In 17th IAPR International Workshop on Document Analysis Systems (DAS 2026), Vienna, Austria, September 2026. [[Code](#)]
- Khadka, S., Baral, S., Tiwari, S., & Ghimire, S. (2025, November). Character Recognition of Nepali Number Plate. In Information Systems for Intelligent Systems, Lecture Notes in Networks and Systems, vol. 1744. Springer. [[Paper](#), [Code](#)]

RESEARCH PROJECTS

Handwritten Devanagari Character Recognition

Capstone Research Project | Pulchowk Campus, IOE, Tribhuvan University

May 2024 – April 2025

- Pioneered a novel, publicly releasable annotated benchmark dataset for handwritten Devanagari **conjoint consonants**, extending the existing character taxonomy and enabling richer linguistic modeling for a historically low-resource script.
- Achieved **state-of-the-art (SOTA) accuracy** in Devanagari handwritten character recognition via transfer learning across deep CNN architectures (ResNet, VGG, EfficientNet), coupled with advanced preprocessing and systematic data-augmentation strategies.
- Conducted rigorous cross-architecture benchmarking to identify optimal model configurations, producing reproducible baselines for future research on Indic scripts.

Character Recognition of Nepali Vehicle Number Plates

Applied Data Science Course | Samsung Innovation Campus

June 2023 – Nov 2024

- Curated and annotated a real-world Nepali license-plate dataset; engineered a two-stage recognition pipeline combining **YOLOv8** for robust plate localization and fine-tuned deep CNNs for Devanagari-script character recognition.
- Applied transfer learning to adapt pre-trained networks, demonstrating consistent performance across varied environmental conditions (illumination, angle, occlusion).

- Results **published in a Springer volume**; awarded **Top Student recognition** by Samsung Innovation Campus (2025).

ArtiFuse: Intelligent Image Inpainting via Generative Deep Learning May 2023 – March 2024
Minor Research Project | Pulchowk Campus, IOE, Tribhuvan University

- Designed and deployed an end-to-end intelligent image-inpainting system using **conditional GANs (Pix2Pix)**, achieving high-fidelity reconstruction of masked and corrupted image regions.
- Integrated the deep learning inference pipeline with a full-stack web application, demonstrating real-time generative model deployment and human-in-the-loop interaction.

INDUSTRY EXPERIENCE

Machine Learning Researcher June 2025 – Present
Digital Age Nepal, Kathmandu, Nepal

- Research into privacy-preserving Machine Learning and interpretability, maintaining data security and compliance while handling sensitive financial information.
- Optimized State-of-the-Art (SOTA) graph networks and unsupervised learning algorithms to maximize predictive performance within strict computational and resource constraints.
- Engineered predictive time-series models for temporal anomaly detection, extracting behavioral insights from large-scale, highly sensitive relational financial databases characterized by complex, high-cardinality features.
- Designed automated information retrieval pipelines for complex legal literature, utilizing Optical Character Recognition (OCR) and Convolutional Neural Networks (CNNs) to accurately extract and process document data.

TECHNICAL SKILLS

Programming Languages	Python, C++
Deep Learning	2D Detection & Segmentation, Pattern Recognition, Transfer Learning, Time-Series Anomaly Detection, Transformers, Privacy-preserving ML
ML / DL Frameworks	PyTorch, TensorFlow, OpenCV
Tools & Infrastructure	FastAPI, LangGraph, Git, Jupyter Notebook
Data & Evaluation	Data Visualization, Statistical Analysis, Benchmarking, Dataset Curation & Annotation

AWARDS & SCHOLARSHIPS

- 2025 **Best Capstone Project Award** — *“Handwritten Devanagari Character Recognition: Conjunct Consonants Dataset Creation and Benchmarking with Transfer Learning,”* Pulchowk Campus, Tribhuvan University (TU).
- 2025 **Top Student Award** — Applied Data Science Course, Samsung Innovation Campus, for the project *“Character Recognition of Nepali Vehicle Number Plate”* at Pulchowk Campus, TU.
- 2024 **Scholarship Recipient** — Fifth Annual Nepal AI School (ANAIS 2024), Nepal Applied Mathematics and Informatics Institute for Research (NAAMII).
- 2025 **Top 12 Nationally** — *TruthShield* (blockchain-based document verification system), SheQual All-Girls Hackathon for Social Cause, SheQual & Ncell, Nepal.

TEACHING EXPERIENCE

2025	Instructor , <i>Internet Literacy & Awareness</i> — Children in Technology Program, LOCUS Nationwide Campaign.
2025	Instructor , <i>Data Science Fundamentals</i> — Software Fellowship, LOCUS, Pulchowk Campus.
2024	Instructor , <i>Advanced Programming Concepts in C</i> — Advanced C Workshop, IT Club, Pulchowk Campus.
2024	Mentor , <i>React JS Intensive</i> — Girls To Code Bootcamp.

INTERNSHIPS & TRAINING

2024	Research Intern , Computer Vision & Deep Learning — <i>Indira Gandhi Delhi Technical University for Women (IGDTUW)</i> , New Delhi, India.
2023	Volunteer Organizer , Hackathon — Fourth Annual Nepal AI School, NAAMII, DIYO.AI & UNDP, Nepal.

COMMUNITY ENGAGEMENT

Event Coverage Head <i>The Zerone</i> — Student Technical Magazine, Pulchowk Campus Led editorial coverage of campus hackathons, technical exhibitions, and national educational tours.	2025
Finance Manager <i>Pulchowk Girls</i> — Women-in-Technology Initiatives, Pulchowk Campus Directed financial planning and budget allocation for women-in-technology empowerment initiatives.	2024